

SUBJECT	Lower Secondary Science Year 8
TOPIC	Metals and Acids
ASSIGNMENT	Five
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: _____

ASSIGNMENT GUIDANCE:

- You should complete this assignment by printing out this sheet and then handwriting in black ink, unless you have obtained specific permission to submit typed work. Then scan it in and submit it, as a single document.
- This is not a timed assignment. We estimate that this assignment may take you between 40 and 50 minutes for the assignment itself. However, if you need to take longer, then please do so.
- Make sure you write your full name clearly in the box at the top of this page.
- In order to complete this assignment, you will need a calculator, ruler, your coursebook and notes.



LS Science Assignment Five

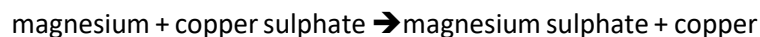
METAL AND ACIDS

1. Part of the reactivity series of metals is shown below.

most reactive	potassium
	sodium
	magnesium
	aluminium
	iron
	lead
least reactive	copper

- a. Dan added a piece of magnesium to a solution of copper sulphate. A displacement reaction took place.

The word equation for the reaction is shown below.



Why is this called a displacement reaction?

(1 mark)

- b. Look at each pair of chemicals in the table below.

Use the reactivity series to predict whether a displacement reaction would take place. Write **yes** or **no** in the second column and give the reason for your decision.

pairs of chemicals	Does a displacement reaction take place? yes or no	reason
iron + sodium chloride		
magnesium + lead nitrate		

(2 marks)





c. Dan wanted to find out where zinc should be placed in the reactivity series.

i. What tests should Dan do to find the correct position of zinc in the reactivity series?

(1 mark)

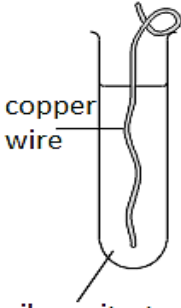
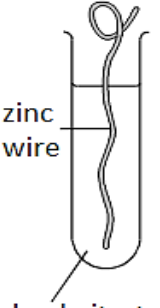
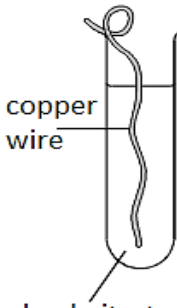
ii. How would Dan use his test results to decide where to put zinc in the reactivity series?

(1 mark)

(Total 5 marks)



2. A group of pupils placed pieces of metal wire in different salt solutions. They recorded their observations in the table below.

experiment	1	2	3
diagram			
observations	crystals of silver formed on the wire	crystals of lead formed on the wire	no change

- a. From these observations, write the order of reactivity of the four metals, copper, lead, silver and zinc.

most reactive _____

least reactive _____.

(2 marks)

- b. The pupils then dipped a new piece of each of the metal wires into dilute hydrochloric acid.

Only **one** of the metals reacted. Which metal was this?

(1 mark)

- c. One pupil predicted that there would be **no** reaction when he put a piece of zinc wire into a solution of silver nitrate.

Was his prediction correct? Explain your answer.

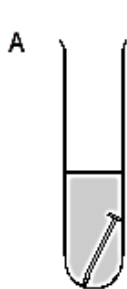
(1 mark)

- d. In nature, gold is never found combined with other elements.

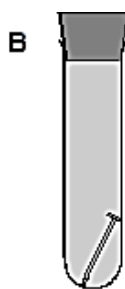
Where should gold be placed in the reactivity series in part (a)? Explain your answer.

(2 marks)
(Total 6 marks)

3. Jessica was investigating the rusting of iron. She set up five experiments as shown below and left the test-tubes for three days.



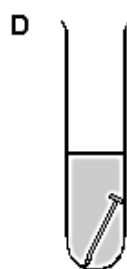
iron nail in distilled water



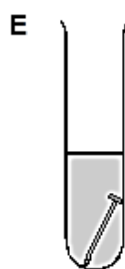
iron nail in tap water
which has been boiled to
remove dissolved gases



iron nail and a chemical
to absorb water vapour



iron nail in sea water



iron nail in vinegar

Jessica wrote the following results in her book.

Test-tube	observation
A	nail slightly rusty
B	nail still shiny
C	nail still shiny
D	nail very rusty
E	nail slightly rusty, bubbles of gas seen

- a. Explain why the nails had **not** rusted in test-tubes B and C.

in test-tube B _____

in test-tube C _____

(2 marks)

- b. In test-tube E the iron nail reacted with the vinegar.

- i. Is vinegar **acidic**, **alkaline** or **neutral**?

_____.

(1 mark)

- ii. When the iron reacted with the vinegar, bubbles of gas were formed.
What gas was formed?

(1 mark)

- c. Before putting the iron nail in test-tube D, Jessica weighed the nail.
After three days she dried and weighed the nail **and** the rust which had formed.

- i. How did the total mass of the nail and rust compare to the mass of the nail at the beginning?

(1 mark)

- ii. Give the reason for your answer.

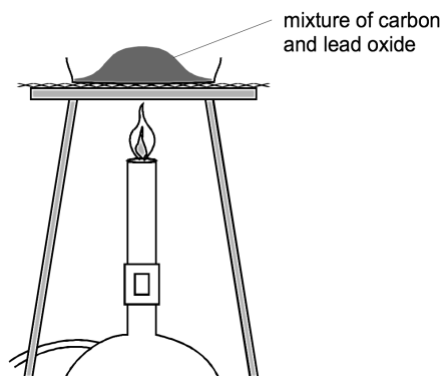
(1 mark)

- d. Jessica concluded that the presence of salt in the water made the nail rust more quickly.
Explain why she drew that conclusion from her experiments.

(1 mark)

(Total 7 marks)

4. Marie mixed 5 g of carbon with 5 g of lead oxide.
She heated the mixture strongly for 15 minutes in a fume cupboard.



After 15 minutes, Marie found some shiny beads in the mixture.

a.

- i. Marie collected all the shiny beads from this experiment.
How could she test them to show they were metal?

(1 mark)

- ii. Marie's test showed that the tiny beads were metal.
What metal were the beads likely to be?

(1 mark)

- b. Marie also expected carbon dioxide to be formed in this experiment.

- i. In carbon dioxide, what element is combined with carbon?

(1 mark)

- ii. Where, apart from the air, did this element come from in this experiment?

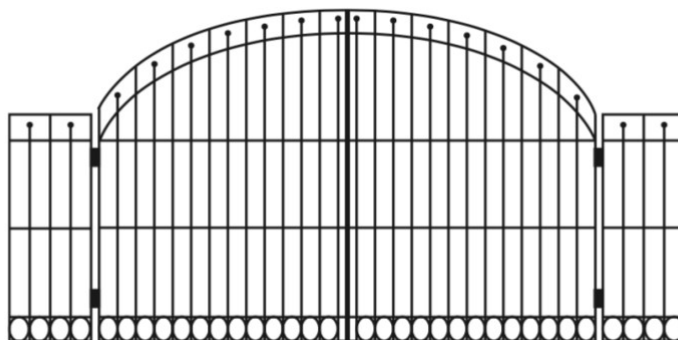
(1 mark)

- c. Give **one** safety precaution Marie should take during this experiment.

(1 mark)

(Total 5 marks)

5. The picture shows a gate made of iron

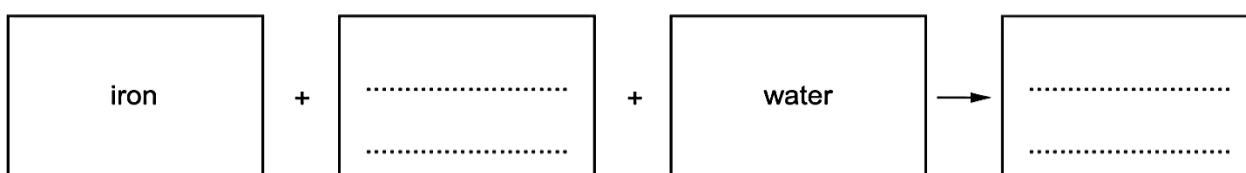


a. Iron has to be painted to stop it rusting. When iron rusts it forms hydrated iron oxide.

i. Where does the oxygen in hydrated iron oxide come from?

(1 mark)

ii. Complete the word equation for rusting.



(2 marks)

iii. Iron oxide is one compound of iron.

Write down the name of **one other** compound of iron.

(1 mark)

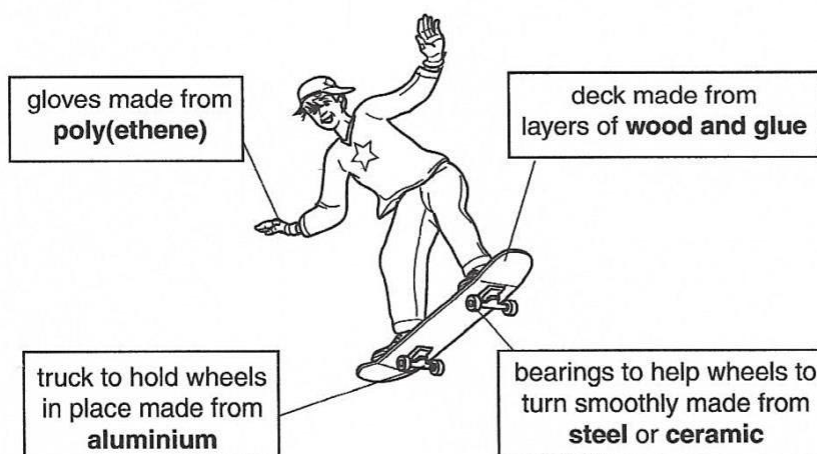
iv. Iron is made of tiny particles that are all the same.

Write down the name of these particles.

(1 mark)

(Total 5 marks)

6. Skateboards and skateboard clothing are made from different materials.



a. From the substances named on the diagram, give the name of:

i. a **polymer**

(1 mark)

ii. an **element** that conducts electricity.

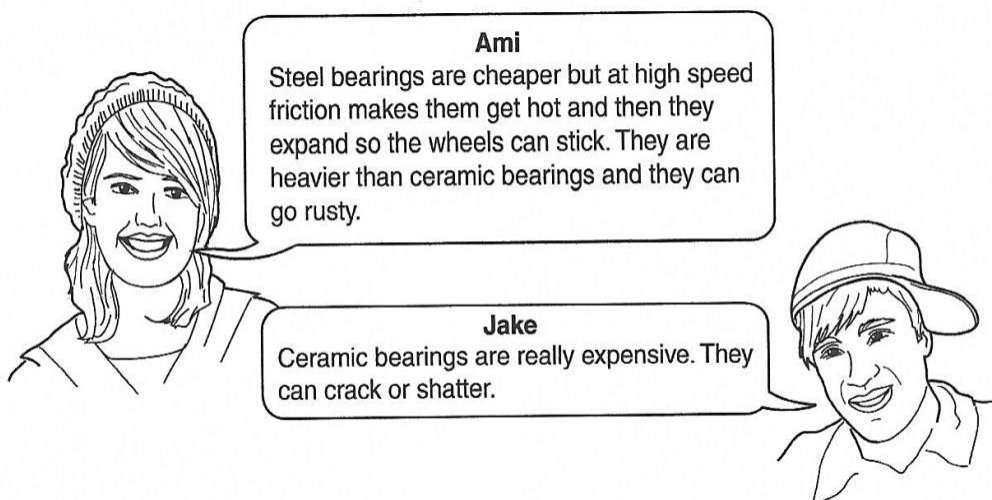
(1 mark)

b. The deck of the skateboard is made of a composite material. Which of the following statements about composite materials are true and which are false? Put a tick in one box in each row.

	true (✓)	false (✓)
all the atoms in a composite are the same		
one example of a composite is glass reinforced plastic		
the properties of a composite are a combination of the properties of the materials used to make it		
materials used to make composites are always arranged in layers		

(2 marks)

- c. Bearings help the wheels on skateboards go round smoothly. Two friends are talking about the disadvantages of steel bearings and ceramic bearings.



- i. Use the information to explain the **advantages** of using ceramic bearings instead of steel.

(2 marks)

- ii. Jake says that ceramics can crack or shatter. What other properties do ceramics have? Put ticks in the boxes next to the **two** correct answers.

can be drawn into wires	☐
conduct electricity	☐
do not melt when heated	☐
flexible	☐
hard	☐

(2 marks)

- iii. One of the ceramics used in skateboards is called Cerbec. Which of the following are also ceramics? Put rings around the **two** correct answers

glass ionic cement melamine nylon

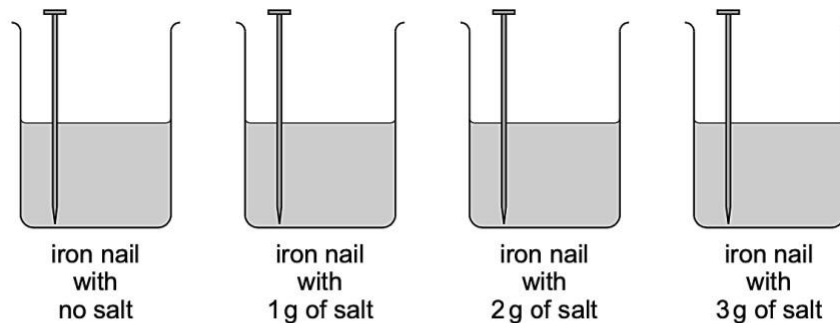
(2 marks)

- d. Ami has an old skateboard. She thinks the materials used to make the skateboard should be recycled. Give one **disadvantage** of recycling the materials in old skateboards.

(1 mark)
(Total 11 marks)

7. Chen investigates rusting.

- He puts distilled water into four beakers.
- He adds different amounts of salt to three of the beakers.
- He puts one iron nail into each beaker.



After one week Chen records his observations.

amount of salt added in g	observation
0	small amount of rust
1	most of the nail in the water is rusty
2	all of the nail in the water is rusty
3	all of the nail in the water is very rusty

a.

- i. Which variable is Chen changing in his investigation?

(1 mark)

- ii. Write down one variable Chen needs to **control**.

(1 mark)



iii. What conclusion can Chen make from his investigation?

(1 mark)

b.

i. Which element dissolved in the water reacts with iron during rusting?

(1 mark)

ii. Chen sets up another beaker.

This time he puts the iron nail in without any water. He then seals the top of the beaker.

Explain why the iron nail in this beaker does not rust.

(1 mark)

(Total 5 marks)

TOTAL FOR ASSIGNMENT 44 MARKS

